

What is the wave in the black-hole scattering project?

Project note

1 The field we evolve

The current project evolves a **massless scalar field on a fixed Schwarzschild spacetime**. The wave is a disturbance in the scalar field. A small perturbation of the metric would instead describe the related problem of gravitational perturbations.

In units $G = c = 1$, the background metric is

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r}. \quad (1)$$

We keep this metric fixed and solve

$$\square_{g_{\text{Schw}}} \varphi = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \varphi) = 0. \quad (2)$$

This is a controlled test-field approximation: if $\Phi = \varepsilon \varphi$, then the scalar stress-energy is quadratic in ε . With no independent gravitational perturbation,

$$T_{\mu\nu}[\Phi] = O(\varepsilon^2), \quad g_{\mu\nu} = g_{\mu\nu}^{\text{Schw}} + O(\varepsilon^2). \quad (3)$$

Thus the scalar can propagate on the background at leading order while its effect on the geometry is neglected. A unit code amplitude is a normalisation of this linear problem.

2 How the three-dimensional wave becomes a radial problem

Separate the angular dependence using a spherical harmonic:

$$\varphi(t, r, \theta, \phi) = \frac{\psi_{\ell m}(t, r)}{r} Y_{\ell m}(\theta, \phi). \quad (4)$$

For a real scalar, use real spherical harmonics or a real combination of complex harmonics. Each mode evolves independently. Define the *tortoise coordinate*

$$x = r_* = r + 2M \ln \left(\frac{r}{2M} - 1 \right). \quad (5)$$

The exterior $2M < r < \infty$ maps to $-\infty < x < \infty$. Suppressing the mode indices, the equation becomes

$$\partial_t^2 \psi = \partial_x^2 \psi - V_\ell(r(x)) \psi, \quad V_\ell(r) = f(r) \left[\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} \right]. \quad (6)$$

This is a one-dimensional wave equation with a spatially varying potential. The potential accounts for angular structure and curvature; it produces scattering outside the horizon. It is not a material surface. For Schwarzschild, the radial potential depends on ℓ but not on m .

3 Specifying the wave means specifying initial data

A second-order equation in time requires **two initial functions**:

$$\psi(0, x) = \psi_0(x), \quad \partial_t \psi(0, x) = \Pi_0(x). \quad (7)$$

The code's two evolved fields will be ψ and $\Pi = \partial_t \psi$. A convenient initial radial profile is a Gaussian:

$$\psi_0(x) = A \exp \left[-\frac{(x - x_0)^2}{2\sigma^2} \right]. \quad (8)$$

Here A is the amplitude, x_0 the centre, and σ the width. The shape alone does not determine which way the wave travels.

Where the potential is negligible, the general solution is

$$\psi(t, x) = F(x - t) + G(x + t). \quad (9)$$

The first term moves towards increasing x ; the second moves towards decreasing x , hence towards the black hole. An initially inward-moving Gaussian therefore uses

$$\Pi_0(x) = +\partial_x \psi_0(x) = -\frac{x - x_0}{\sigma^2} \psi_0(x). \quad (10)$$

The minus sign choice, $\Pi_0 = -\partial_x \psi_0$, selects outward motion. Choosing $\Pi_0 = 0$ gives equal inward and outward parts in the free-wave limit. At finite radius, a nonzero potential means these are local free-wave prescriptions, not exact pure scattering states.

A single angular mode is a radial pulse with an angular pattern: $\ell = 0$ is spherically symmetric, while $\ell = 2$ has quadrupolar structure. A wave packet localised on just one side of the black hole would require a superposition of angular modes.

4 Properties you can vary

Parameter	Effect in the experiment
A	Scales the field linearly and its energy quadratically; does not change the background's ringdown frequencies.
σ	Changes the pulse width and frequency content. A narrower Gaussian has a broader frequency spectrum.
x_0	Changes the launch position, arrival time and initial overlap with the scattering potential.
Π_0	Selects initial inward motion, outward motion, or a mixture.
ℓ	Changes the angular structure, potential and ringdown spectrum.
k_0	Adds an oscillation scale to a Gaussian wave packet.

For the last option, use

$$\psi_0(x) = A e^{-(x-x_0)^2/(2\sigma^2)} \cos[k_0(x - x_0)], \quad \Pi_0 = \partial_x \psi_0. \quad (11)$$

Far from the potential, k_0 sets the approximate carrier frequency. Varying it lets you study how the reflected and transmitted energy fractions depend on frequency.

5 How to see the propagation

Once time evolution is implemented, three complementary views will show what the field does:

1. **Snapshots or an animation of $\psi(t, x)$.** Watch the initial pulse move inward, interact with the potential, and split into inward and outward components.
2. **A spacetime heat map of $\psi(t, x)$.** Put x on the horizontal axis, t on the vertical axis, and field value in colour. In regions where the potential is small, propagation follows $x - t = \text{constant}$ or $x + t = \text{constant}$.
3. **A waveform at a fixed observer.** Record

$$w(t) = \psi(t, x_{\text{obs}}). \quad (12)$$

An observer outside the barrier can see the outgoing scattered signal and its subsequent decaying oscillations, called *ringdown*.

The one-dimensional plots show the reduced radial field. To reconstruct the scalar at a spacetime point, restore the angular factor and $1/r$:

$$\varphi(t, r, \theta, \phi) = \frac{\psi(t, r_*(r))}{r} Y_{\ell m}(\theta, \phi). \quad (13)$$

The waveform's time coordinate is Schwarzschild t ; a static observer at radius r_{obs} has proper time $d\tau = \sqrt{f(r_{\text{obs}})} dt$.

The transmitted part moves towards $x \rightarrow -\infty$, corresponding to the horizon. In the planned finite numerical domain, the first experiment uses distant reflecting endpoints and a time window chosen to exclude significant returning boundary reflections. A larger-domain comparison will check that the measured scattering is independent of those walls.

6 Connection to gravitational waves and the project goal

If instead we perturb the metric,

$$g_{\mu\nu} = g_{\mu\nu}^{\text{Schw}} + \varepsilon h_{\mu\nu} + O(\varepsilon^2), \quad (14)$$

the leading vacuum problem is the linearised Einstein equation $\delta G_{\mu\nu}[h] = 0$. Schwarzschild gravitational perturbations can be described by Regge–Wheeler and Zerilli master equations of the form

$$\partial_t^2 \Psi = \partial_x^2 \Psi - V_{\text{grav}}(r) \Psi. \quad (15)$$

The numerical structure is similar, but the variables and potentials have different physical meanings. This offers a natural later extension; the present scalar waveform is not a gravitational-wave strain.

The current goal is to build and validate a C++/AMReX solver for scalar scattering, extract its ringdown, and study numerical accuracy and adaptive refinement. The first practical exercise remains a flat, periodic travelling wave with $V = 0$, before introducing Schwarzschild curvature.

Further reading: E. Berti, V. Cardoso and A. O. Starinets, *Quasinormal modes of black holes and black branes*, [arXiv:0905.2975](https://arxiv.org/abs/0905.2975). The repository's longer theory guide contains the detailed derivations.